AI for Transportation: From concepts to implementation

Prof. Jinhua Zhao | Professor of Cities and Transportation

AI for Transportation

L4: Generative AI

S1: AI to Understand Bus Operators' Preferences

S2: AI for Generative Urban Design

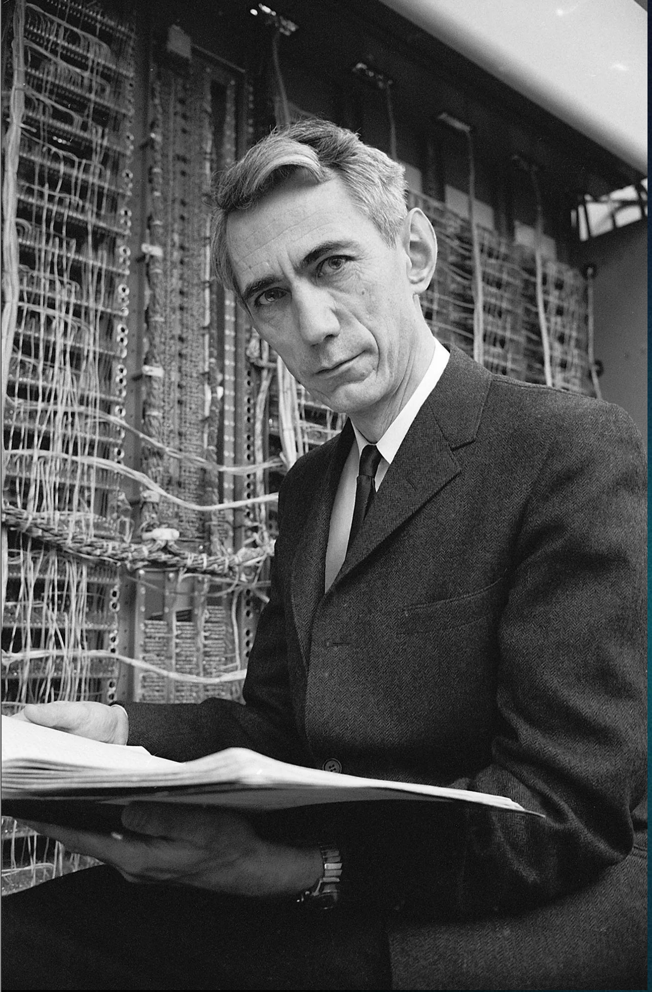
S3: AI for Mobility Trajectory Generation

S4: Outlook

Discriminative AI

VS

Generative AI



Discriminative

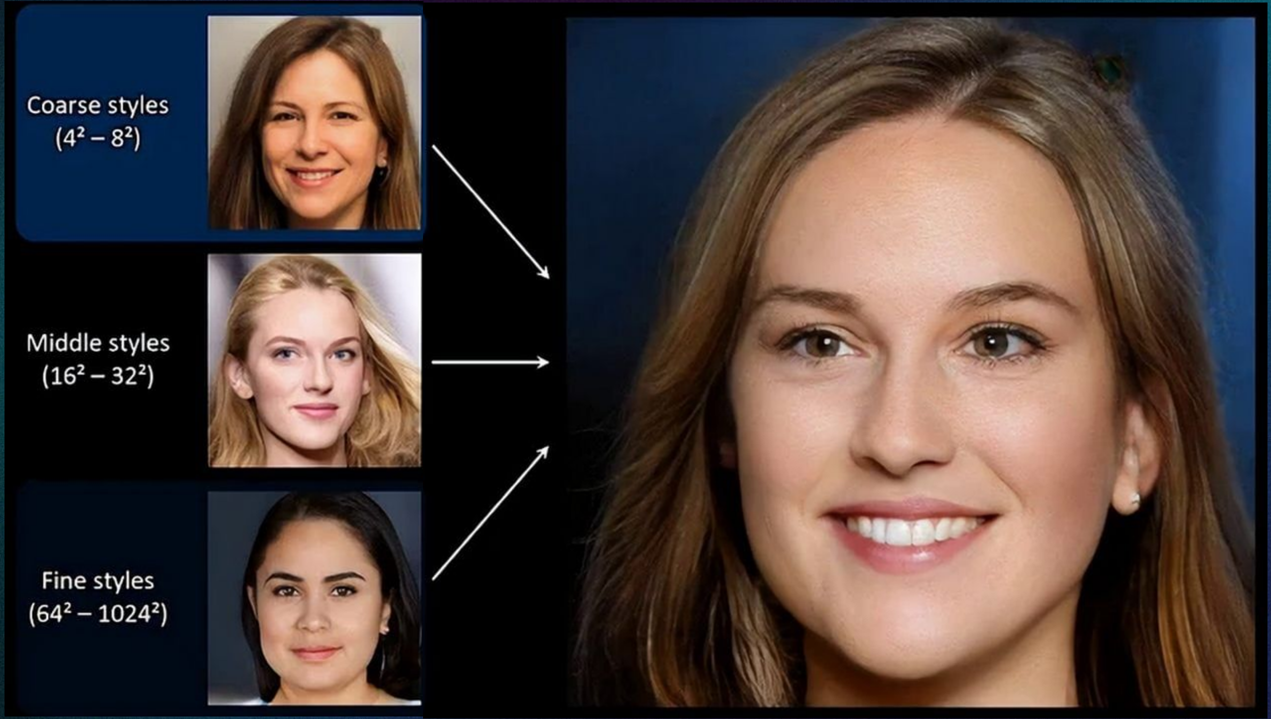
Claude Shannon

Generative

Face shape, Pose, hair

Facial features, eyes

Color scheme



Discriminative vs. Generative AI

Discriminative Models:

- Models the **conditional** probability of the target variable Y, given the observed variables X
- Ex. Given trip time, trip distance, trip-maker’s age, income, car ownership, what mode will this person take?
- Logistic regression, decision trees, support vector machines, some neural networks, etc.

Generative Models:

- Models the **joint** probability between the observed variable X and the target variable Y
- Ex. What is the probability of a specific combination of trip time, trip distance, trip-maker’s age, income, car ownership occurring?
- Hidden Markov models, Gaussian mixture models, variational autoencoders, diffusion models, etc.

Main difference:

- Generative models have a more **global** view of the process, and can **generate new data**.

Four Sample Applications

Collect

AI to Understand Bus Operators' Preferences

Amelia Baum, Jiangbo Yu, John Attanucci, Haris Koutsopoulos & Jinhua Zhao with CTA

Work | Interface

Create

AI for Generative Urban Design

Qingyi Wang, Shenhao Wang, Mingyi He, Yuebing Liang & Jinhua Zhao

Work

Generate

AI for Mobility Trajectory Generation

Baoshen Guo & Jinhua Zhao with M3S

Sense

Infer

AI for Causal Inference with LLMs

Ao Qu & Jinhua Zhao with M3S

Assist

Create

Interface

Sense

AI to Understand Bus Drivers' Preferences

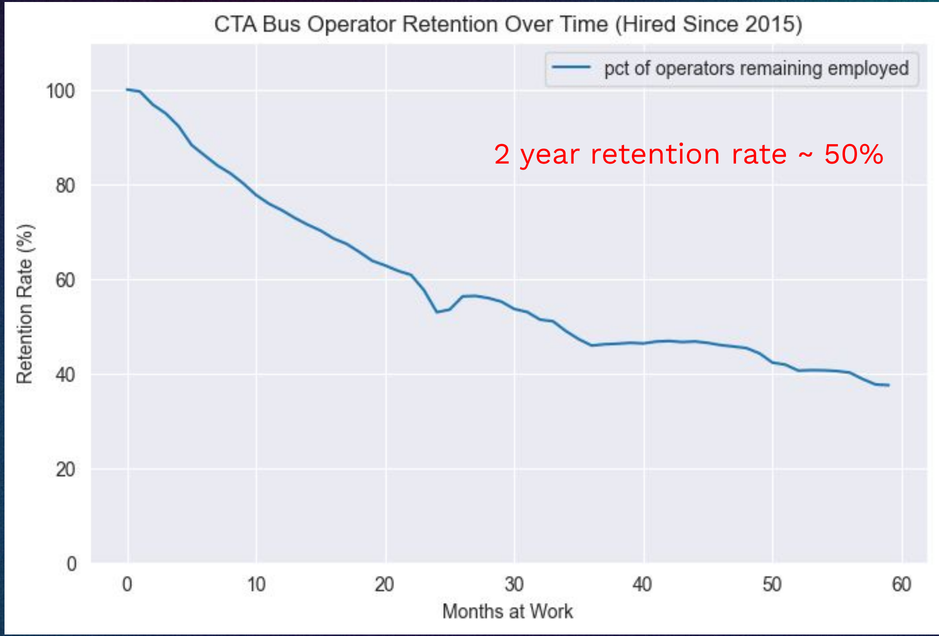
Amelia Baum, Jiangbo Yu, John Attanucci, Haris Koutsopoulos & Jinhua Zhao

Retention of Bus Operators

Retention of transit operators is a major issue

Work assignment practices lack flexibility and disadvantage newer operators.

Little past work on leveraging scheduling practices to improve operator retention and working conditions

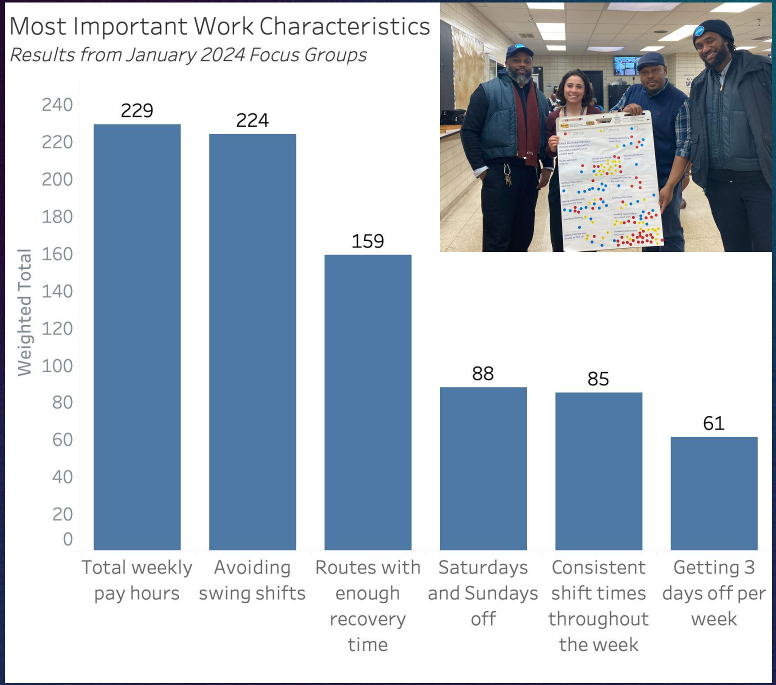


Understand driver preferences

Understand the work preferences of bus operators at the CTA

Evaluate the potential of scheduling innovations such as the increased utilization of block runs and rostering

Increase retention rates and quality of life of bus operators

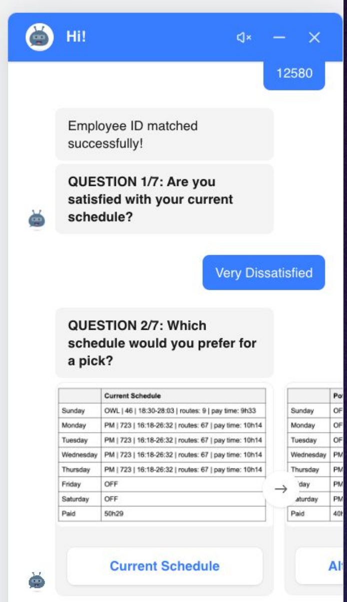


Stated Preference Survey With Real World Alternatives

Goals

Assess CTA bus operators' willingness to work new 4-day block schedules using realistic options aligned with seniority.

Collect qualitative, unstructured feedback about what operators think of these schedules

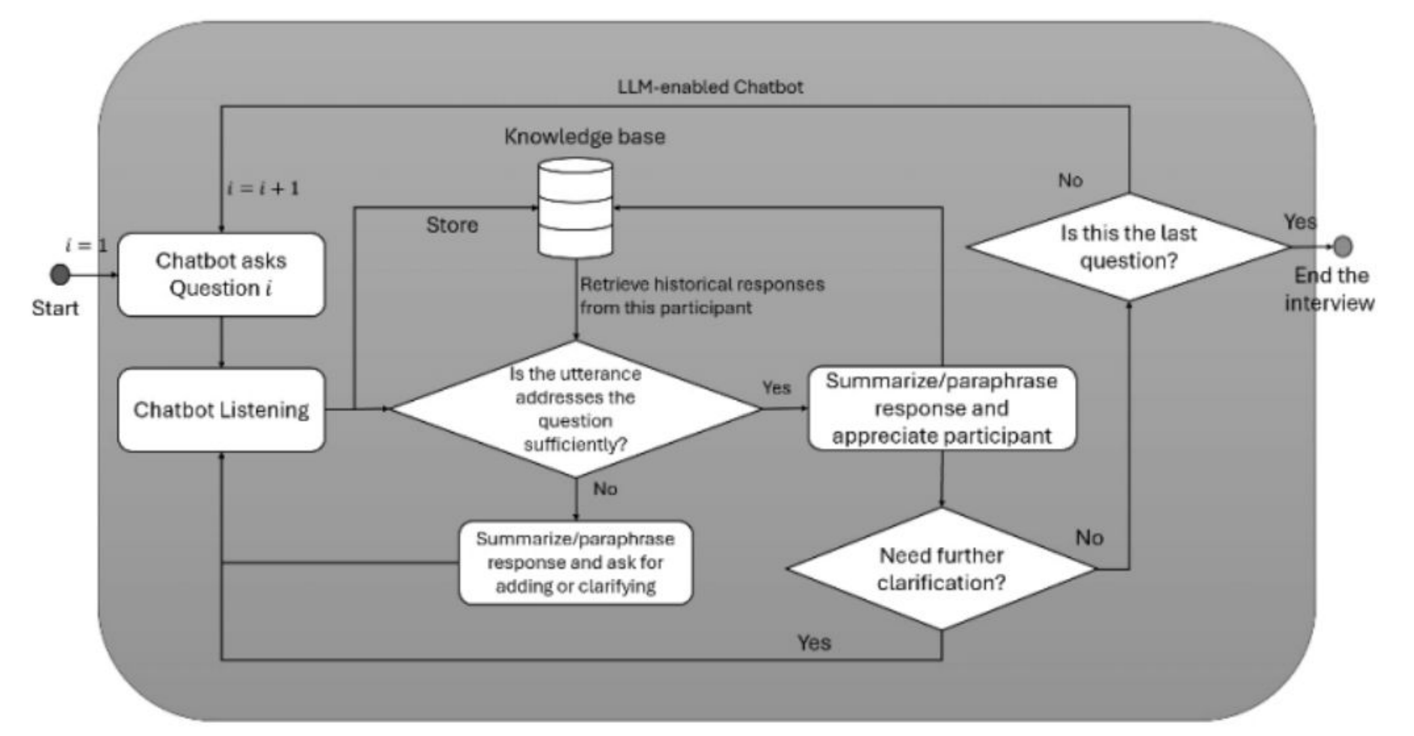


Questionnaire Surveys

VS

Semi-structured interviews

Conversational AI Agent
To enable semi-structured interviews



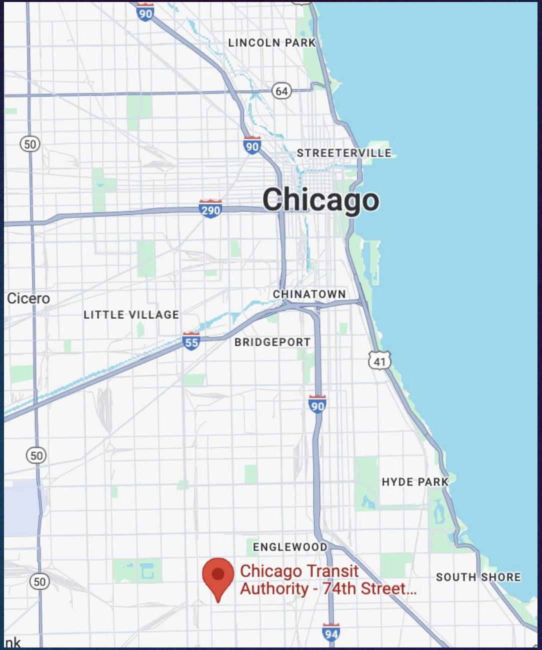
Implementation & Logistics

~20 hours in the 74th street garage over 4 days (during the spring 2025 pick March 8-11)

Operators used their personal phones or CTA- provided tablets to complete the survey

Operators were given a $20 Amazon gift card for participating

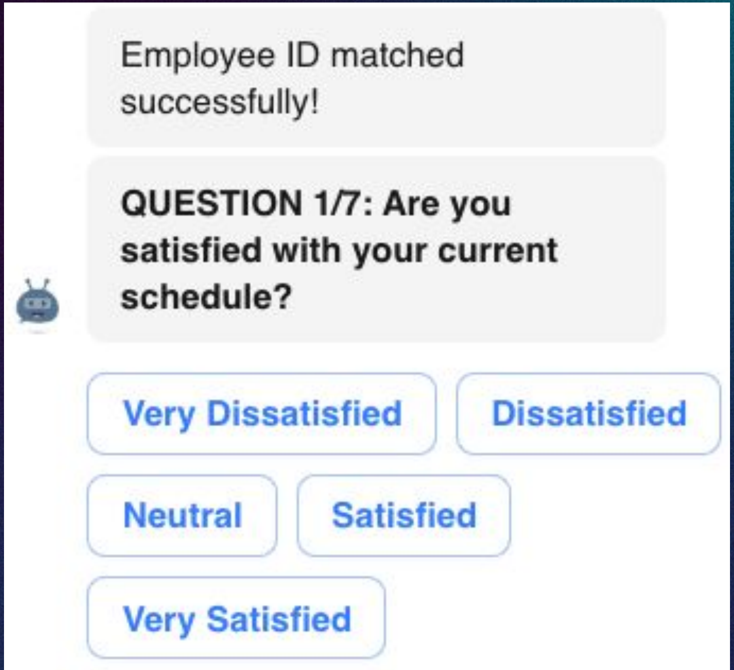
Survey took <10 minutes to complete for most operators



Survey Structure

Part 1: Current Schedule

Satisfaction Respondents are asked their level of satisfaction with their current schedule (very unsatisfied to very satisfied)



Survey Structure

Part 2: Paired-Choice Questions:

Respondents compare their actual work schedule to an alternative generated by a weighted distance algorithm, ensuring similarity in some aspects but variation in others.

Which Schedule would you prefer for a pick?

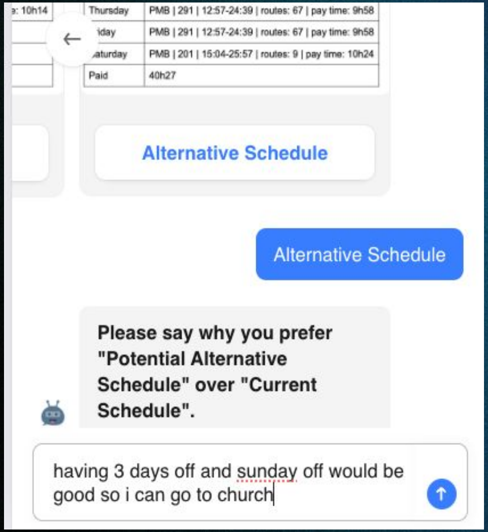
	Current Schedule
Sunday	PM | 213 | 8:24-19:01 | routes: 62 | pay time: 10h14
Monday	PM | 218 | 12:12-21:16 | routes: 62 | pay time: 8h19
Tuesday	PM | 218 | 12:12-21:16 | routes: 62 | pay time: 8h19
Wednesday	OFF
Thursday	PM | 218 | 12:12-21:16 | routes: 62 | pay time: 8h19
Friday	OFF
Saturday	PM | 629 | 7:05-18:34 | routes: 59,63W,55N,62H | pay time: 11h00
Paid	46h11

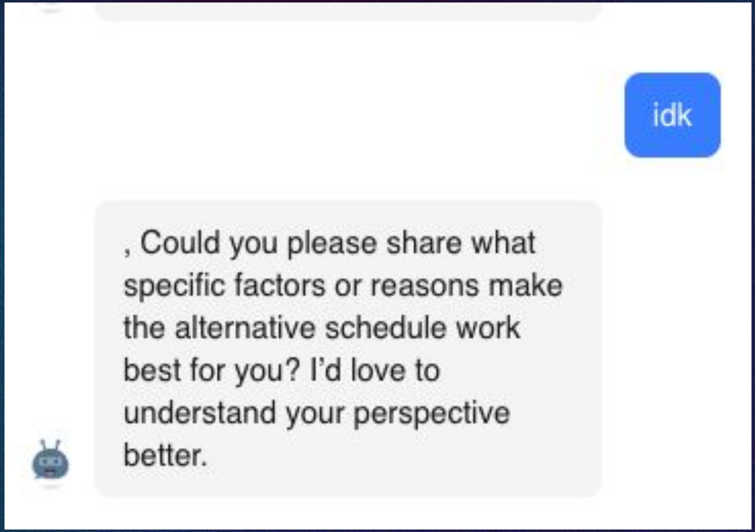
	Potential Schedule Alternative
Sunday	OFF
Monday	OFF
Tuesday	OFF
Wednesday	SWB | 151 | 5:49-18:49 | routes: 59, 63W, 67 | pay time: 10h40
Thursday	PMB | 237 | 7:57-20:57 | routes: 94 | pay time: 11h16
Friday	PMB | 237 | 7:57-20:57 | routes: 94 | pay time: 11h16
Saturday	PMB | 129 | 8:11-21:08 | routes: 59, 62H, 55N, 63W, 9 | pay time: 11h28
Paid	47h00

Survey Structure

Part 3: Dynamic Follow-Ups:

A chatbot asks open-ended follow-up questions, interpreting responses in real time and prompting for clarification dynamically.



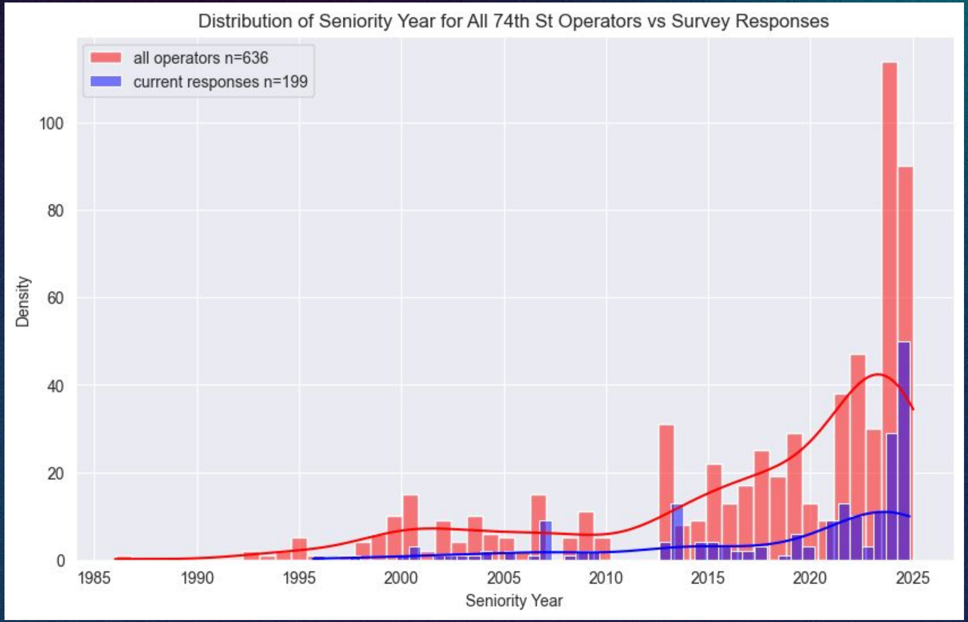


Survey Participation

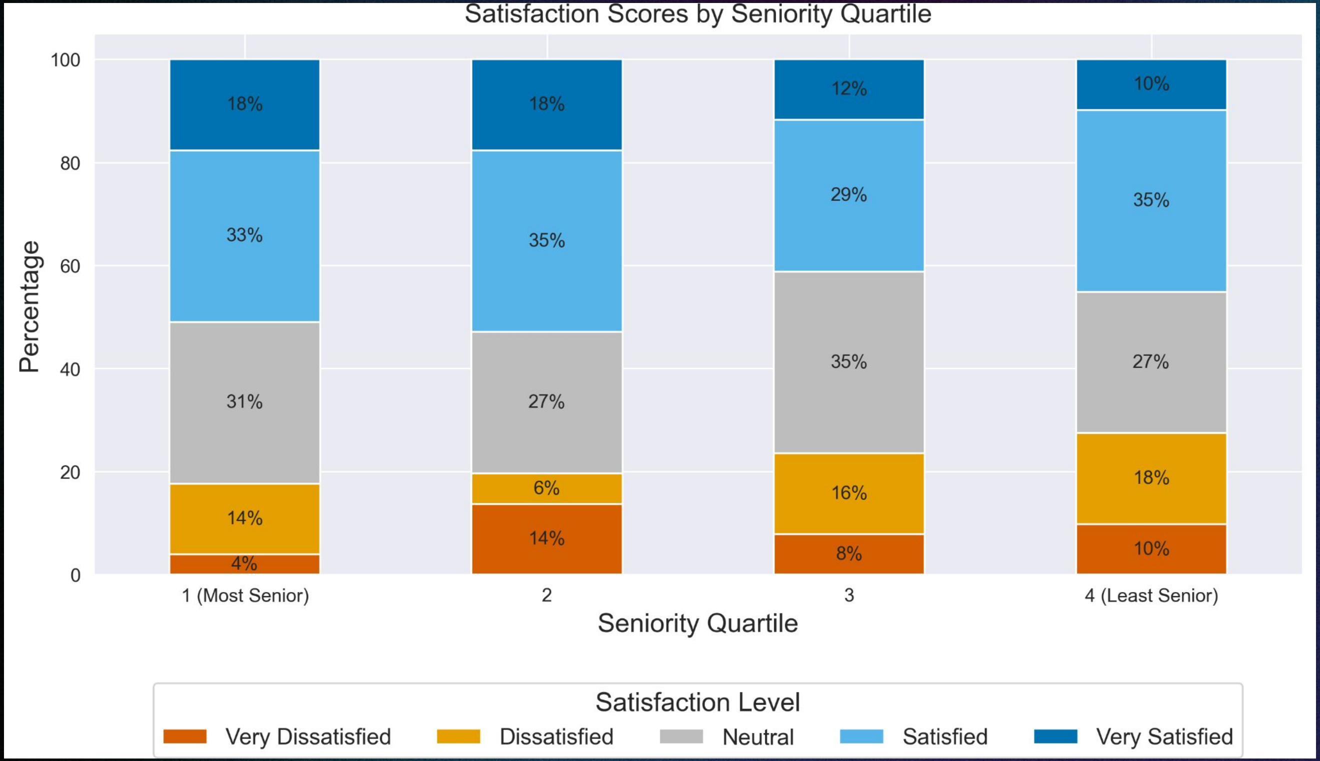
204 respondents total

201 compensated (some responses trickled in from the app after the garage visit)

199 with seniority date records accessible



Key Results – Current Schedule Satisfaction



n=204

Proportion of Operators Choosing Alternative Schedule

Seniority Quartile*	Current Non Block Operators	Current Block Operators	Num Respondents
1 (most senior)	10%	13%	138
2	17%	35%	141
3	39%	N/A	105
4 (least senior)	65%	N/A	84

*quartiles are determined from the full population of operators at the garage, thus the number of responses per quartile are standard.

LLM Aided Thematic Analysis of Free Responses

The alternative schedule has streets mixed in it that I do not work

Block schedule is good. I like 3 days off.

More hours

I like late pm shifts

Like having Mondays off

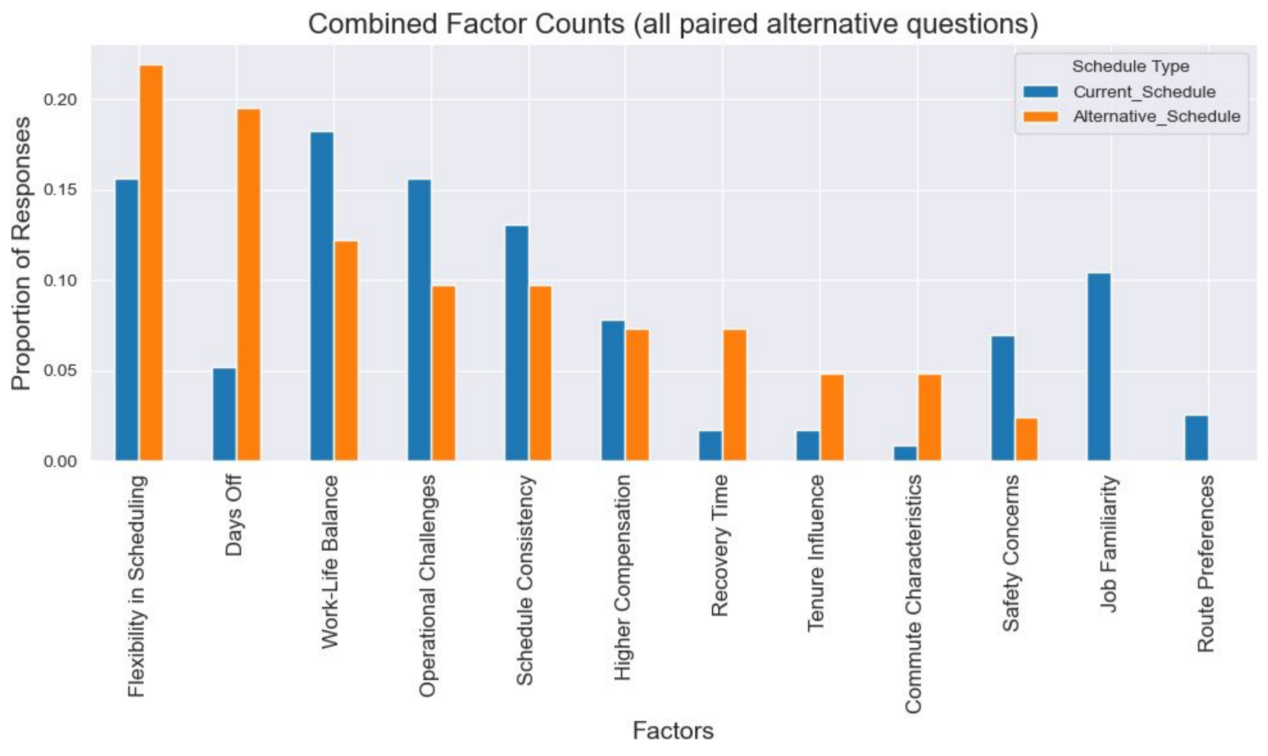
My schedule times to get off are better

Again, my current schedule because I picked for my mental. Certain streets I just would rather not

I prefer current schedule because it's paying a few mins less but you don't have to work as hard and I like having 2 off days instead of 3

I like the hours

Don't want to do 94



More money less stress current schedule was great I feel safer keeping my bus on fb

The time works better for me. I have a better quality of life. It took me well over 20 years to get this time.

I'm a day person

Because by working early you don't dill with a whole lot

Because Sunday off is the most important day for me again because I'm active in my church. I'm a deacon, sing on the praise team and also apart of the security force.

All of my appointments mainly be thursdays

Less driving time for overall health thus more time for exercise & preparation.

Key Takeaways

199* operators surveys are representative seniority-wise

Strong engagement and thoughtful feedback

Overall, ~30% preferred the matched alternative over their current schedule, but ~72% would likely be satisfied with the hybrid rostering paradigm

Senior and block operators least likely to switch, but >60% of newest operators preferred proposed schedules

Pay was a key factor, some operators expressed that block runs were too long

**204 operators total, measured seniority for 199*

AI acquires language capacity

Create

Work

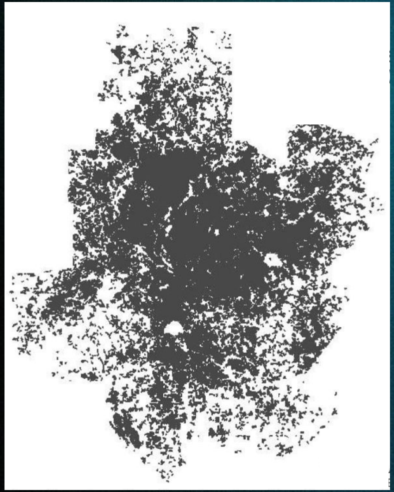
AI for Generative Urban Design

Qingyi Wang, Shenhao Wang, Mingyi He, Yuebing Liang & Jinhua Zhao

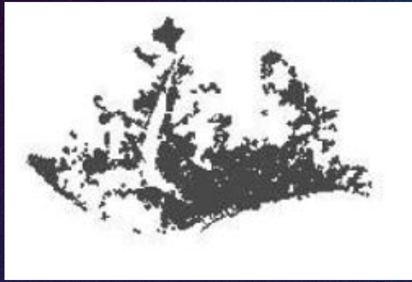
The New York High Line

Decarbonize Mobility: Urban structure is the starting point

Atlanta, USA
Population: 2.5 million (1990)
Built-up Area: 4,280 sq. km.



Barcelona, Spain
Population: 2.8 million (1990)
Built-up Area: 162 sq. km.



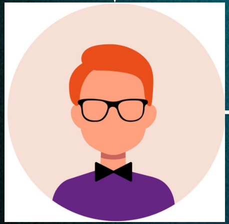
(15 minute city)

Source: Alain Bertaud and Harry Richardson, Transit and Density: Atlanta, the United States and Western Europe. https://www.researchgate.net/publication/248151171_Transit_and_Density_Atlanta_the_United_States_and_Western_Europe/download

Typical planning process

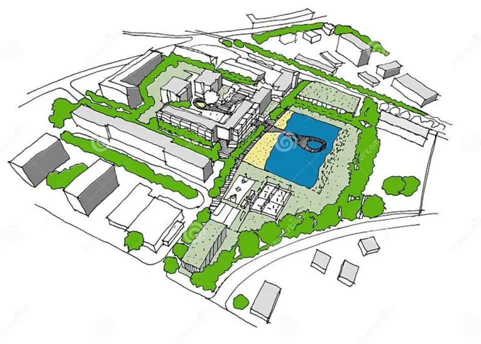
Zoning, Policy, Traffic, Social interactions, Natural Environment, Infrastructure, Public Engagement, ……

Feedback and Iterate (weeks and months)



Urban Planner

Generate plans, layouts and visualizations



https://www.123rf.com/photo_16135751_sketch-of-a-new-development-urban-idea.html

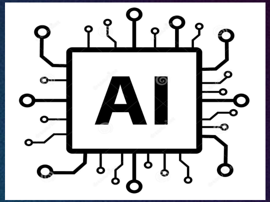
The human-machine planning collaboration

Zoning, Policy, Traffic, Social interactions...

Urban Planner

Human Guidance

Give me a design in Chicago with 70% apartments, 10% park, 10% retail, and a lot of green space!



Generative AI

Generate Designs

Feedback and Iterate (minutes)

What do planners want?

An Example

A piece of land 35 mile north of downtown Chicago

A target land use

- 35% residential
- 15% commercial
- 10% open parking

What do planners want?

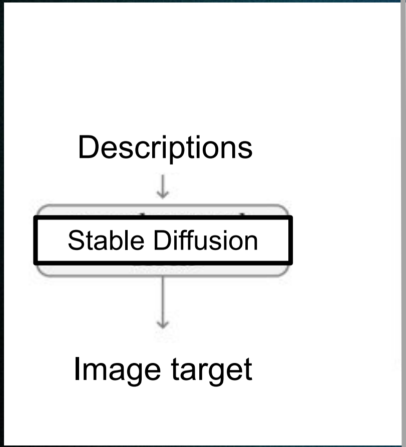
Expectations of Generative Planning Tools

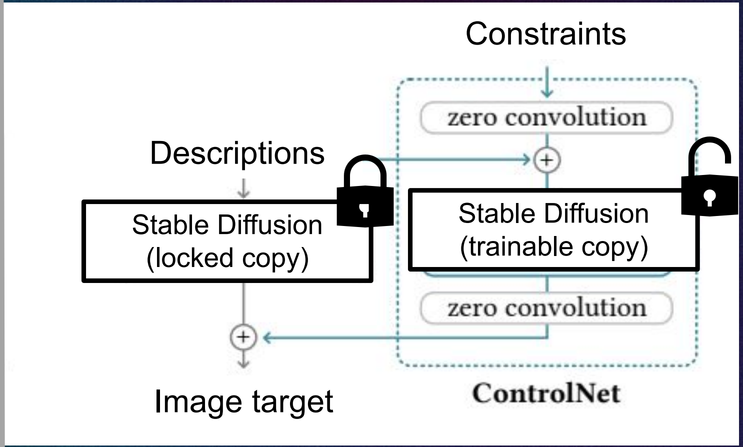
- Having control over AI-generated land-use patterns
- Respecting existing infrastructure and natural environment
- Learn and apply different urban textures

Can A.I. do it?

Fine-tuning stable diffusion using ControlNet

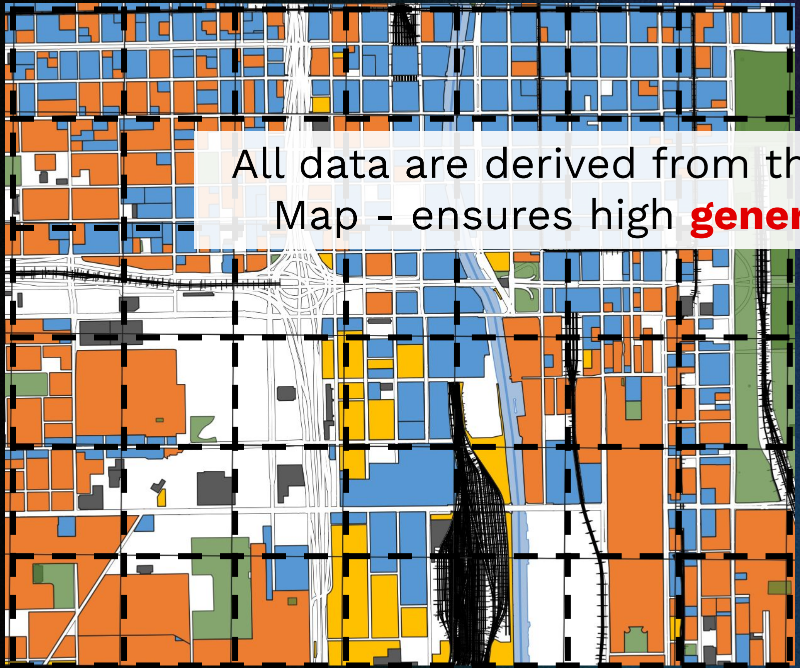
Stable Diffusion | Stable Diffusion with ControlNet





Data: Open Street Map

Land use patterns



Existing infrastructure and natural environment

All data are derived from the publicly available Open Street Map - ensures high **generalizability** and **reproducibility**

Residential | Commercial | Park | Industrial | Open parking | Farmland

A piece of land 35 mile north of downtown Chicago

A target land use

- 35% residential
- 15% commercial
- 10% open parking

A piece of land 35 mile north of downtown Chicago

A target land use

- 65% industrial,
- 20% commercial,
- 5% open parking

A piece of land 35 mile north of downtown Chicago

A target land use

- 100% residential

1. Having **control** over AI-generated land-use patterns

35% residential,
15% commercial,
10% open parking

65% industrial,
20% commercial,
5% open parking

100% residential

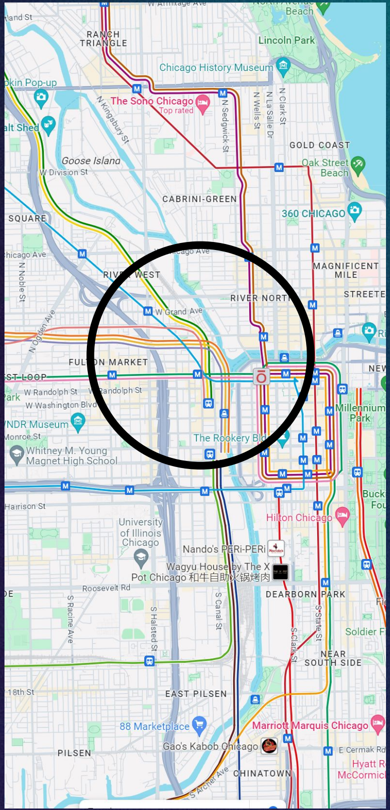
Respecting existing infrastructure and natural environment

Descriptions

City in Chicago.
35% residential
20% commercial,
5% open parking.
Mix of houses and apartments.
Dense building coverage.

Constraints

Generated Aerial Imagery



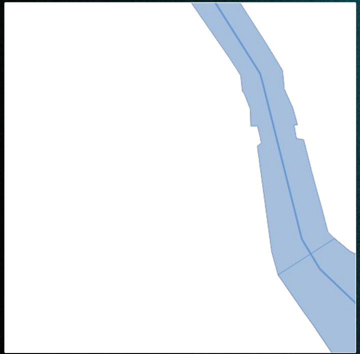
Respecting existing infrastructure and natural environment

Add waterways

Descriptions

City in Chicago.
35% residential
20% commercial,
5% open parking.
Mix of houses and apartments.
Dense building coverage.

Constraints



Generated Aerial Imagery

Respecting existing infrastructure and natural environment

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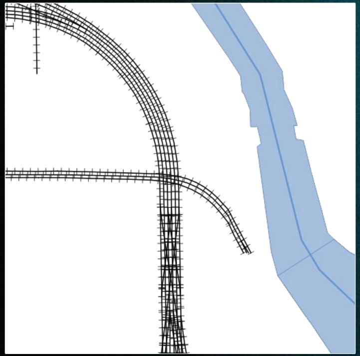
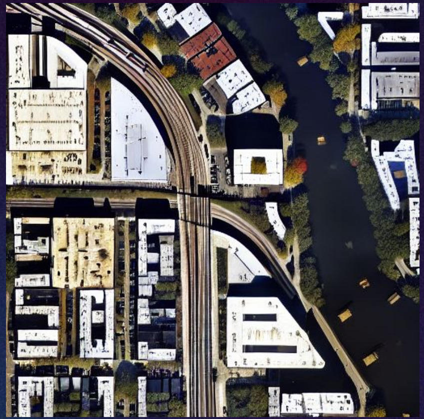


Add railways



Generated Aerial Imagery

Respecting existing infrastructure and natural environment

Descriptions

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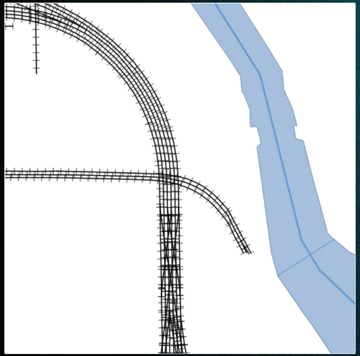


Add roads

Generated Aerial Imagery

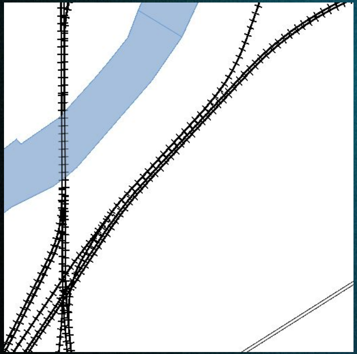
Generated Aerial Imagery

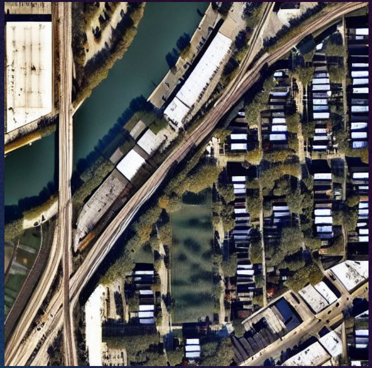
Learn and apply different urban textures

Descriptions

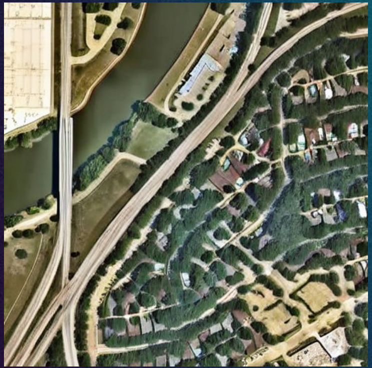
City in {city name}.
40% residential
15% industrial,
15% commercial,
10% park,
5% open parking.
All houses.
Medium building coverage.

Constraints

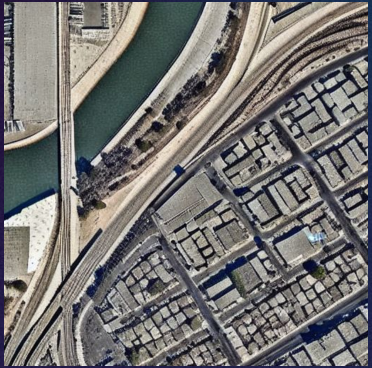




Chicago



Dallas



Los Angeles

2023-12-05

What do planners want?

Expectations of Generative Planning Tools

- Having control over AI-generated land-use patterns
- Respecting existing infrastructure and natural environment
- Learn and apply different urban textures

Learn and apply different urban textures

Descriptions

City in {city name}.
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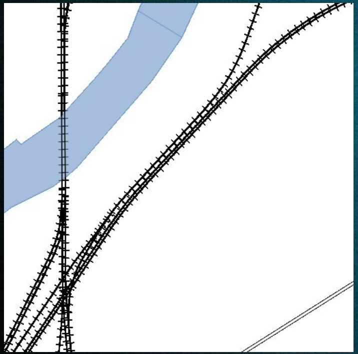
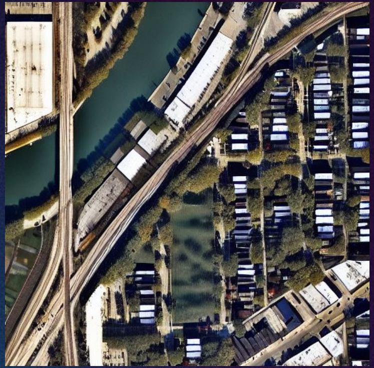
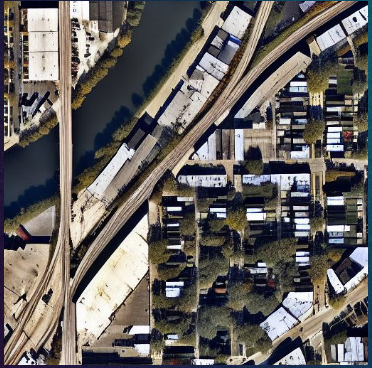
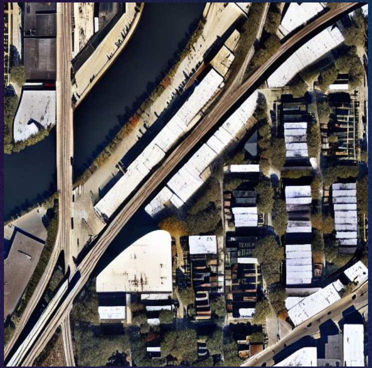




Design 1



Design 2



Design 3

2023-12-05

What will future planners do?

Immediacy changes the dialog

Planners have to be more responsive

A.I. enables the public

Planning is a tool to allocate power

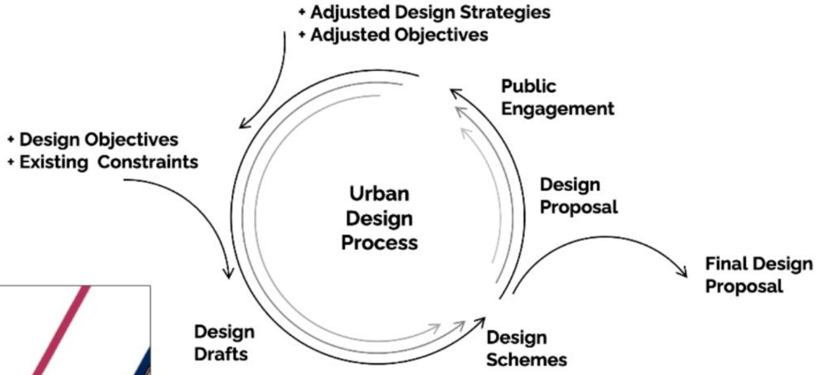
Essense of planners' job

Design creator → Design critique
Design creator → Design communicator
Design creator → Design negotiator

How do planners and AI work together?

Background / **Generative Urban Design**

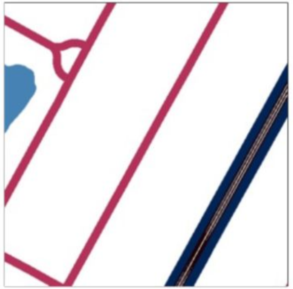
Urban design is iterative / labor-intensive / time-consuming...



Evaluation on AI adherence to human design loop ?

Alignment with practical, real-world needs ?

Intricate design principles ?



Input:
Site constraints

AI

Human Expertise x **Diffusion Models**

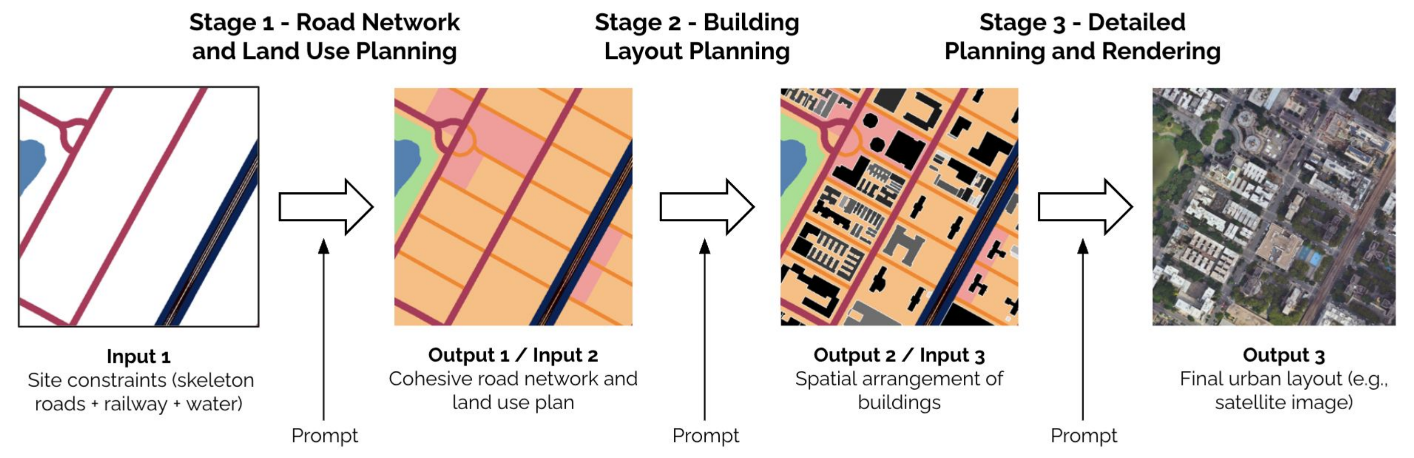
Output:
Final urban layout

Human-Centered AI for Urban Design

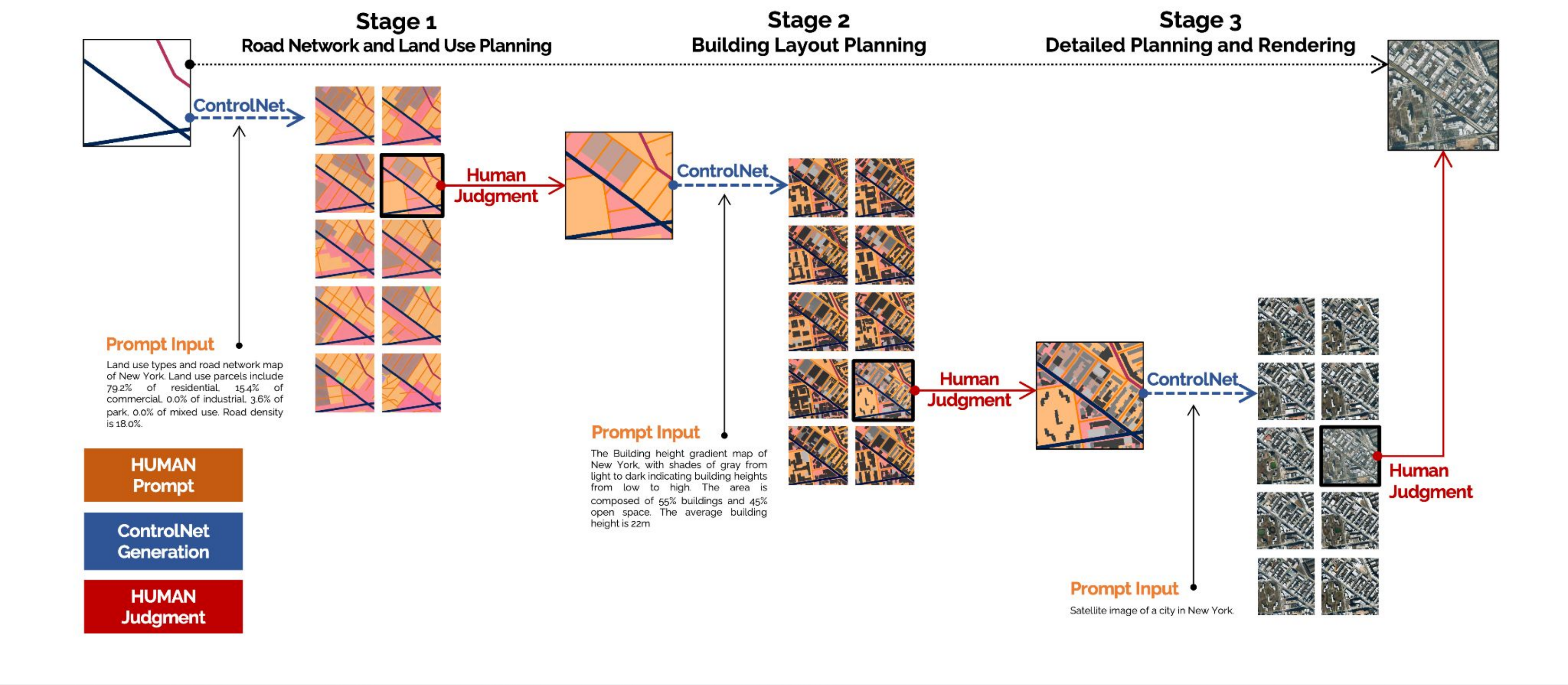
A Stepwise Approach Integrating Human Expertise with AI Models

Human-in-the-loop Design Process:

Human designers maintain control over the design process, enabling iterative adjustments and refinements at each stage.



Human-Centered AI for Urban Design



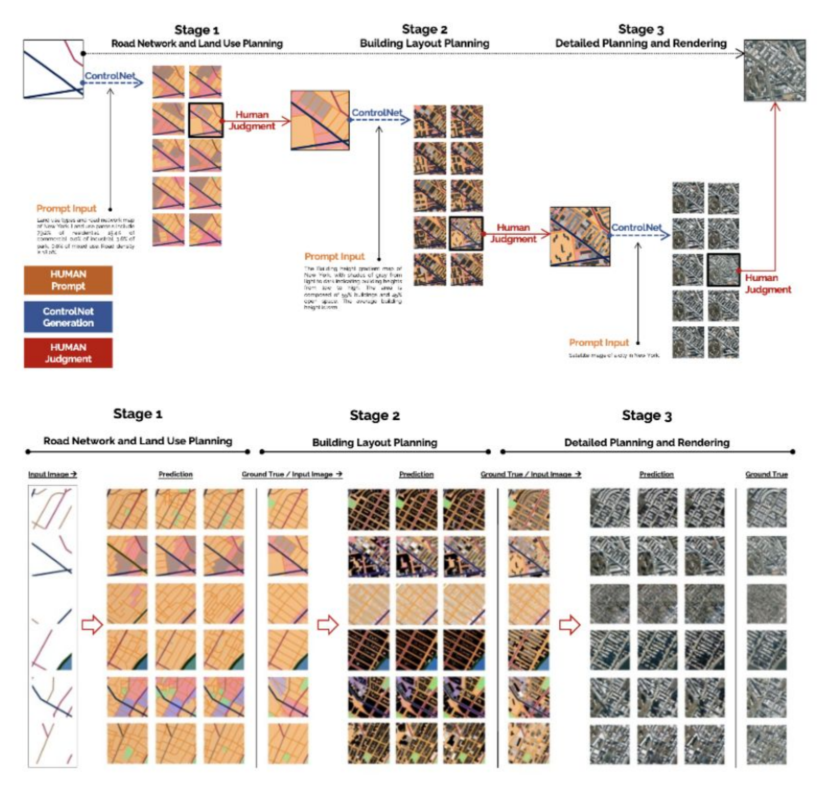
Main Findings

1. **Model Effectiveness**
   - ControlNet understands designer prompts and generates effective urban layouts, **outperforming the baseline model in both image fidelity and prompt alignment**.

2. **Stepwise Control Advantage**
   - Stepwise generation is superior to end-to-end in terms of **human control, output quality, and alignment with instructions**.
   - An Example of Neuro-symbolic AI.

3. **Collaborative Design**
   - **Human-Centered AI** for Urban Design enhances diversity and human control, promoting efficiency in urban design and facilitating public communication and iterative refinement.

4. **Design Pattern Discovery**
   - Through cross-city comparisons and the application of city model transfer, the model uncovers **hidden urban design patterns.**



Generate | Sense

AI for Mobility Trajectory Generation

Baoshen Guo, Shenhao Wang & Jinhua Zhao

Link to the slide:

AI for Mobility Trajectory Generation

https://docs.google.com/presentation/d/1sefj7AB2vqEYkcBBembAMBIVSVnysnEPPTfa0ycpN04

S4: **Outlook**

Three Sample Applications

Collect

AI to Understand Bus Operators' Preferences

Amelia Baum, Jiangbo Yu, John Attanucci, Haris Koutsopoulos & Jinhua Zhao with CTA

Work Interface

Create

AI for Generative Urban Design

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Work

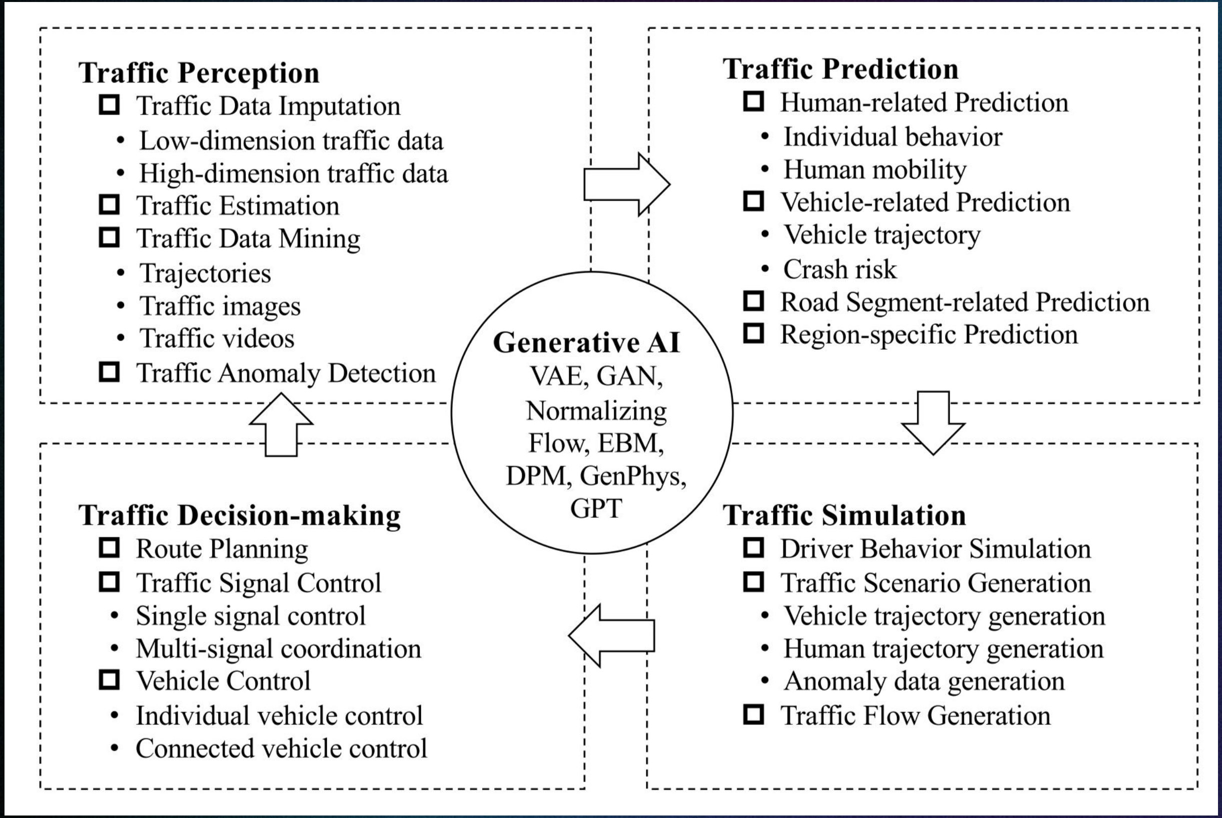
Generate

AI for Mobility Trajectory Generation

Baoshen Guo & Jinhua Zhao with M3S

Sense

Generative AI in intelligent transportation systems



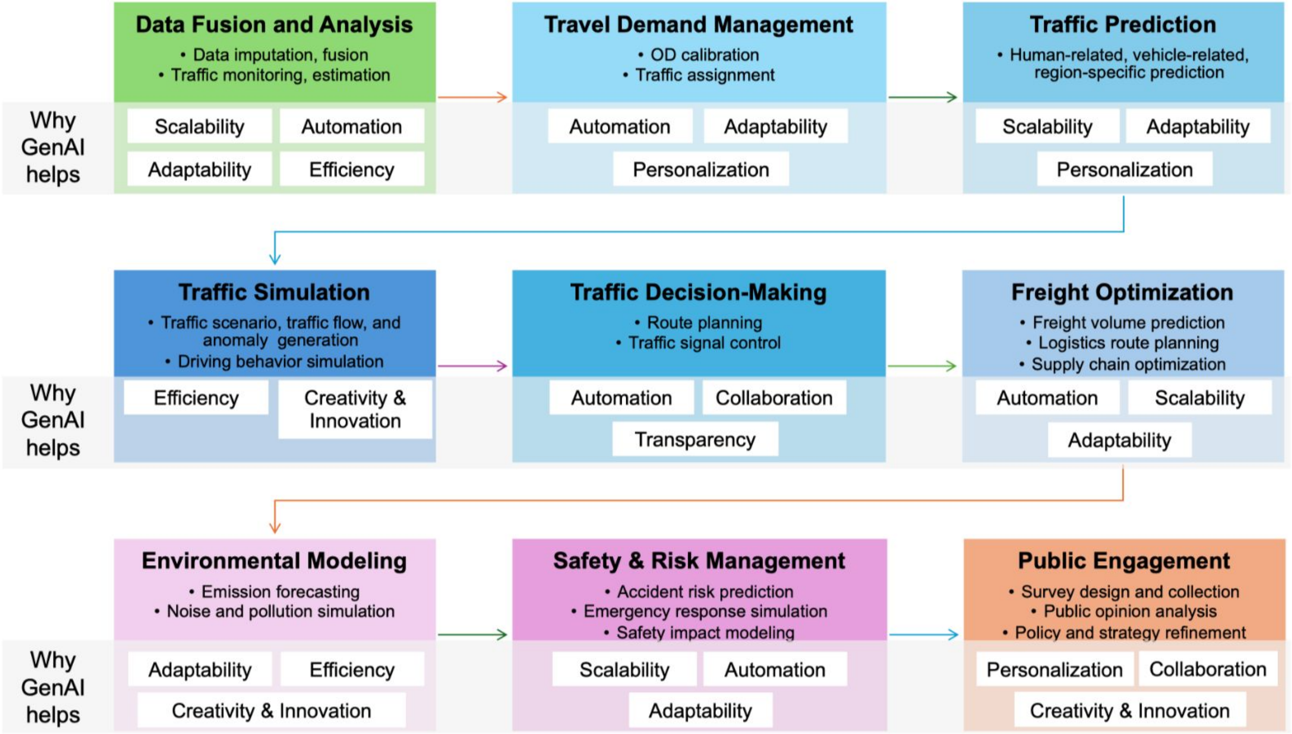
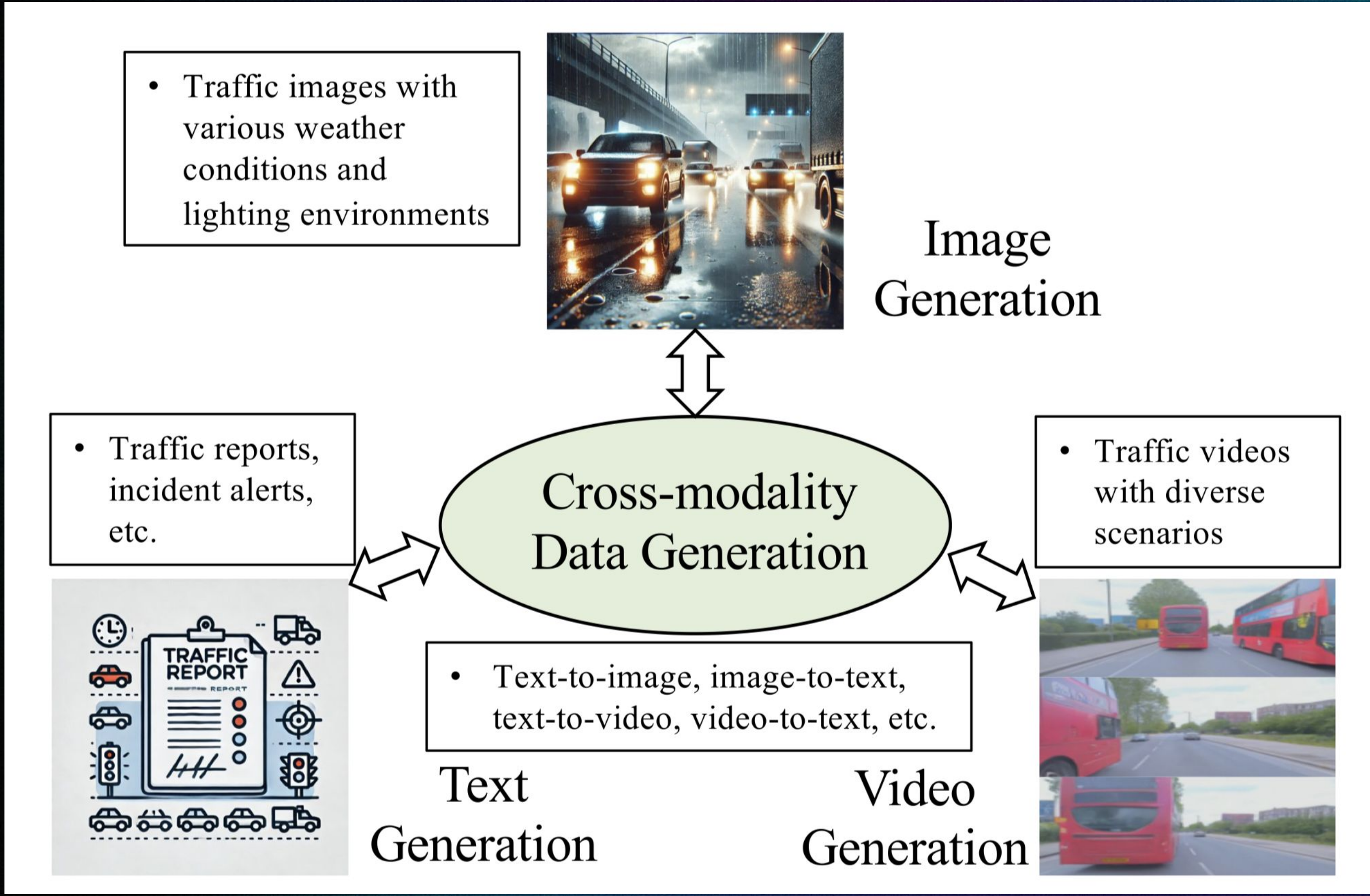


Figure 3: Tasks in transportation planning and potentials of Generative AI for these tasks. A *partial* list of reasons on why GenAI helps transportation planning is also included.


• Traffic images with various weather conditions and lighting environments
Image Generation
• Traffic reports, incident alerts, etc.
Cross-modality Data Generation
• Traffic videos with diverse scenarios
TRAFFIC REPORT
• Text-to-image, image-to-text, text-to-video, video-to-text, etc.
Text Generation
Video Generation

Value	Debates	Mitigation Plans
Scalability	GenAI models can handle vast amounts of data across large geographic areas, allowing for comprehensive analysis and decision-making. The adoption of generative AI requires significant investment in training, data acquisition, and system integration, which can be costly for many DOTs.	• Invest in cloud computing and distributed processing solutions to ensure affordable scalability. • Prioritize models that are efficient in terms of computation.
Automation	Automation can reduce human error and free up planners to focus on more strategic decisions, improving overall efficiency. Excessive automation may lead to job displacement and reduce the oversight needed in decision-making.	• Implement hybrid models where GenAI assists planners but does not fully replace human decision-making, ensuring accountability.
Adaptability	GenAI models can quickly adapt to changing conditions, optimizing traffic flow and route planning in real-time. GenAI may not fully account for highly dynamic, unpredictable scenarios, leading to potential system failures in unexpected situations.	• Continuously update and train models with real-time data to improve their adaptability to new and unforeseen conditions.
Efficiency	GenAI can optimize transportation networks and decision-making, leading to better resource allocation and reduced congestion. GenAI can optimize transportation networks and decision-making processes, leading to better resource allocation and reduced congestion.	• Ensure robust data validation and cleaning protocols before input into models • Incorporate redundancy in model design for error correction.
Personalization	Personalization of routes, services, and communication for diverse user groups, leading to improved user satisfaction and efficiency. Personalized services may inadvertently exclude underrepresented or marginalized groups, exacerbating existing disparities.	• Focus on inclusivity and fairness in AI design, ensuring that models are trained on diverse, representative datasets and that equity considerations are built into the process.
Transparency	GenAI can self-explain, provide clear, interpretable outputs that can be easily understood by planners and stakeholders, enhancing decision-making processes. GenAI models, particularly black-box models, can lack transparency, making it difficult for stakeholders to trust or understand the inner mechanism of the model and the decisions generated by the model.	• Use interpretable machine learning techniques, such as explainable AI (XAI), to provide insights into how decisions are made and ensure clarity in AI models' outputs.
Creativity & Innovation	GenAI fosters innovation by creating new solutions to complex transportation problems, such as optimizing traffic flow, predicting accidents, and designing infrastructure. over-reliance on AI-generated solutions may stifle human creativity and lead to overly standardized solutions.	• Ensure that AI is used as a tool to augment human creativity rather than replace it. • Encourage human-AI collaboration and maintain human oversight in critical decision-making areas.

Figure 12: Debates over the potentials of GenAI in transportation planning and their mitigations

Conclusion

Generative AI: A new paradigm

1. **Text, images, audio, and video**
   a. Currently most promising approaches: Transformer- and Diffusion-based
2. **Directly applicable to urban planning and transportation**
   a. Imagining new environments
   b. Processing big data
   c. Creating transportation data
   d. Knowledge production